Last updated July 8, 2024

Urban Retail Water Supplier - Water Conservation, Supply, and Demand

The State Water Board collects monthly data on water use and actions undertaken by water suppliers, e.g. water agencies that deliver 3,000 annual acre-feet of water and/or have 3,000 or more service connections. Data collection began in 2014 and has been collected on a monthly basis since then. Data corresponding to the June 2014-June 2023 period were collected through the DRINC Portal; data collection changed to the SAFER Clearinghouse in 2023 when the DRINC Portal was retired.

Dataset parameter	**DRINC Portal**	**SAFER Clearinghouse**	**Per-Supplier Water Supply and Demand**	**Conservation Actions**
Timespan	June 2014-June 2023	January 2023 onwards	June 2014 onwards (June 2014-December 2022 DRINC data, January 2023 onwards Clearinghouse data)	October 2020-June 2023
Reporting Entity	Water Supplier	Water System	Aggregated Water System	Water Supplier
Water Data Units	Both original units and gallons	Both original units and gallons	Gallons only	N/A

Dataset releases, previously located on the Board's Water Conservation Portal, are now released on the Open Data Portal. The full dataset containing supplier-specific volumetric data (combining data collected from DRINC, for June 2014-December 2022, and from the SAFER Clearinghouse, for January 2023 onwards) is available as a CSV file with a corresponding data API.

The corresponding volumetric data from the SAFER Clearinghouse, reported at the individual water system level, is also available for January 2023 onwards, in the original reported units as well as gallons.

Reported conservation activities are broken out into separate tables.

The conservation actions dataset is provided in two formats: the listed subcategory version, in which actions are reported as comma-separated lists of activities under subcategories such as demand reduction and water use restrictions; and the per-column version, in which each action has its own column, with a "Y" if the supplier indicated the action was implemented by the supplier in the reporting month.

Data Quality Flags

The Water Conservation and Production Report contains fields to indicate aspects of data quality. These "Flag" columns are informed by the corresponding "Modified Z-score" fields, which make it easier to spot outliers. Quantities for which the corresponding flag column are labeled "Flagged"

are likely misreported; these quantities are excluded from numbers that Water Board staff present on the Tableau dashboard or any other Board presentations that use this dataset. Quantities that are flagged may be updated by reporters at any time. (Reports submitted for January 2023 onwards can be directly edited in the SAFER Clearinghouse. Reports submitted prior to January 2023 must be corrected by Water Board staff; please contact orpp-monthly-conservation@waterboards.ca.gov to submit corrections.)

The following sections explain the process for calculating the modified z-scores and associated flags.

Modified Z-Score

The z-score is a statistical measurement that can be used to determine if a value might be an outlier in a dataset. This value is defined as the number of standard deviations from the mean of the dataset, assuming certain things about the data generating process. The larger the magnitude of the z-score, the further the value is from the mean; therefore, z-scores with a magnitude beyond a certain threshold (often 3) are considered to be outliers.

Water production and population data do not necessarily follow a normal distribution, which means that both the mean and standard deviation can be sensitive to anomalies. The modified z-score adjusts for this sensitivity by using medians instead of means.

Modified z-score equation

The equation for the modified z-score is

$$z_i = \frac{x_i - \tilde{X}}{k \times MAD}$$

where

- z_i is the modified z-score
- x_i is the data point that is being compared to the full data array x
- $\tilde{X}$ is the median of the data array x
- k is a statistical correction factor, in this case 1.4826
- MAD is the median absolute deviation, where

$$MAD = median(|x_i - median(x)|)$$

Calculating the modified z-score columns

Each data point for which there is a corresponding modified z-score is compared to other values that were reported by the same supplier, within a three-month window. For example, residential water use for Supplier X in December 2023 will be compared to all other residential water use values reported by Supplier X for November, December, and January in all available reporting years. The monthly groups can generally be defined as the month of comparison, as well as the month before and after. Therefore,

- January (1) will be compared to all December (12), January (1), and February (2)
- February (2) will be compared to all January (1), February (2), and March (3)
- March (3) will be compared to all February (2), March (3), and April (4)
- etc.

The algorithm to calculate the modified z-score can generally be defined as follows:

```
For each supplier X:
      subset_1 = dataset[supplier=X]
      for mon in range (1,12):
            if mon == 1:
                  months=[12,1,2]
            else if mon == 12:
                  months=[11,12,1]
            else:
                  months=[mon-1,mon,mon+1]
            subset_2=subset_1[months][variable_list]
            if length(subset_2)>5:
                  for var in variable_list:
                        mod_z_var = mod_z_score_calc(subset_2[var])
            data_final=data_final.append(subset_2[month=mon])
```

Data Flags and Criteria

Modified z-scores with a magnitude of 3.5 or more are considered to be outliers. The threshold to generate a data flag is a modified z-score with a magnitude of 7 or more in the dataset, in order to avoid flagging values associated with extreme weather (drought/atmospheric rivers), which might lead to larger or smaller values than average.

Modified z-score values are available for the following variables:

- Potable Supply Minus Sold Gallons
- Potable Demand Residential Gallons
- R-GPCD
- Potable Supply Minus Sold Minus Ag Gallons

The criteria for the data flags are as follows:

- Potable Supply Minus Sold Flag: Potable Supply Minus Sold Gallons Modified Z-Score has a magnitude of 7 or more
- Residential Flag: Potable Demand Residential Gallons Modified Z-Score has a magnitude of 7 or more **and/or** R-GPCD is less than 20
- Potable Supply Minus Sold Minus Ag Flag: Potable Supply Minus Sold Minus Ag Gallons Modified Z-Score has a magnitude of 7 or more

Data Dictionaries

The data dictionaries for each of the volumetric datasets are in the following section. The conservation actions data dictionaries are forthcoming in a future version of the documentation.

Data dictionary for Per-Supplier Water Supply and Demand dataset

Per-Supplier Water Supply and Demand Dataset Field Name	Label	Type	Description
ORG_ID	Organization ID	Text	DWR-assigned identification number associated with the urban retail water supplier
SUPPLIER_NAME	Supplier Name	Text	Name of the urban retail water supplier
WATER_SYSTEM_ ID	Water System ID	Text	Identification number(s) associated with the water system(s) in the supplier service area
COUNTY	County	Text	County/counties encompassing the supplier service area
HYDRO_REGION	Hydrologic Region	Text	Name of the principal hydrologic region encompassing the supplier's service area
CLIMATE_ZONE	Climate Zone	Text	Name of the principal climate zone encompassing the supplier's service area
REPORT_PERIOD_ START_DATE	Reporting Period Start Date	Timestamp	Calendar date corresponding to the start of the reporting period
REPORT_PERIOD_ END_DATE	Reporting Period End Date	Timestamp	Calendar date corresponding to the end of the reporting period
POP_REPORT_ PERIOD	Population During Reporting Period	Numeric	Estimate of the number of permanent residents served potable water during the reporting month
DWR_STANDARD_ LEVEL	DWR State Standard Level corresponding to Stage	Text	A standardized numeric level (0-6) corresponding to a supplier's local Water Shortage Contingency Plan stage
DWR_STANDARD_ LEVEL_FLAG	DWR State Standard Level Flag	Text	If "Reported as no Water Shortage Contingency Plan", supplier indicated that they do not have a Water Shortage Contingency Plan - hence "Not Applicable" for the DWR State Standard Level
POTABLE_SUPPLY_ GAL	Potable Supply Gallons	Numeric	Available for January 2023 report onwards (Clearinghouse only). Total volume of potable water produced from groundwater, GWUDI, surface water, springs, or external sources.

Per-Supplier Water Supply and Demand Dataset Field Name	Label	Type	Description
POTABLE_SUPPLY_PRELIM_EST	Potable Supply Gallons Preliminary Estimate	Text	If "Yes", supplier indicated that the reported potable supply number is a preliminary value
POTABLE_VOL_SOLD_GAL	Potable Volume Sold or Delivered Gallons	Numeric	Total potable volume, in gallons, delivered from one supplier to other entities
POTABLE_SUPPLY_MINUS_SOLD_GAL	Potable Supply Minus Sold Gallons	Numeric	June 2014-December 2022 reports (DRINC dataset): reported. January 2023 onwards (Clearinghouse dataset): calculated difference. Total potable supply, excluding potable water delivered to other entities, in gallons
POTABLE_SUPPLY_MINUS_SOLD_FLAG	Potable Supply Minus Sold Gallons Flag	Text	If "Flagged", indicates that the corresponding Potable Supply Minus Sold Gallons number is an outlier in the seasonal subset of values, corresponding to a modified Z-score of +/-7 or more (see Potable Supply Minus Sold Gallons Modified Z-Score columns)
POTABLE_DEMAND_RES_GAL	Potable Demand Residential Gallons	Numeric	June 2014-December 2022 reports (DRINC dataset): derived from reported percent residential use and reported potable supply. January 2023 onwards (Clearinghouse dataset): reported. Total potable volume, in gallons, of single and multi-family water use.
R-GPCD	R-GPCD	Numeric	Calculated field. Residential daily per capita water use, calculated as (residential gallons)/((number of days in reporting period)×(population during reporting period))
POTABLE_DEMAND_RES_PRELIM_EST	Potable Demand Residential Gallons Preliminary Estimate	Text	If "Yes", supplier indicated that the reported residential volumes are preliminary values
RES_FLAG	Residential Flag	Text	If "Flagged", indicates that 1) the corresponding Potable Demand Residential Gallons number is an outlier in the seasonal subset of values, corresponding to a modified Z-score of ±7 or more (see Potable Demand Residential Gallons Modified Z-Score); and/or 2) the calculated R-GPCD was less than 20.

Per-Supplier Water Supply and Demand Dataset Field Name	Label	Type	Description
POTABLE_DEMAND_AG_GAL	Potable Demand Agriculture Gallons	Numeric	Total potable volume, in gallons, of water used to irrigate commercially grown crops, nurseries, etc.
POTABLE_DEMAND_CII_IRR_GAL	Potable Demand Commercial, Industrial, and Institutional (With Landscape Irrigation) Gallons	Numeric	June 2014-December 2022 reports (DRINC dataset): reported field (voluntary data). January 2023 onwards (Clearinghouse dataset): calculated sum of Potable Demand Commercial, Industrial, and Institutional Gallons and Potable Demand Non-Residential Metered Irrigation Gallons. Potable volume of commercial, industrial, and institutional water use, including irrigation of parks, medians, etc.
POTABLE_DEMAND_CII_GAL	Potable Demand Commercial, Industrial, and Institutional Gallons	Numeric	Available for January 2023 report onwards (Clearinghouse only). Potable volume of commercial, industrial, and institutional (CII) water use. Does NOT include irrigation of parks, medians, etc if said irrigation is associated with dedicated irrigation meters (DIMs).
POTABLE_DEMAND_IRR_GAL	Potable Demand Non-Residential Metered Irrigation Gallons	Numeric	Available for January 2023 report onwards (Clearinghouse only). Potable volume of CII irrigation (parks, medians, etc) associated with DIMs.
POTABLE_DEMAND_O_GAL	Potable Demand Other Gallons	Numeric	Available for January 2023 report onwards (Clearinghouse only). Potable volume of water used for purposes other than the previously reported columns. May include fire suppression, street cleaning, line flushing, construction/temporary meters, etc.
POTABLE_DEMAND_PRELIM_EST	Potable Demand Gallons Preliminary Estimate	Text	If "Yes", supplier indicated that some or all of the reported potable demand fields are preliminary values

Per-Supplier Water Supply and Demand Dataset Field Name	Label	Type	Description
RECYCLED_DEMAND_GAL	Recycled Water Demand Gallons	Numeric	June 2014-December 2022 reports (DRINC dataset): reported field (voluntary data). January 2023 onwards (Clearinghouse dataset): calculated sum of reported residential and non-residential recycled water use. Total volume of recycled water used by the supplier for both residential and non-residential purposes during the reporting period
NON_POTABLE_DEMAND_PRELIM_EST	Non-Potable Demand Gallons Preliminary Estimate	Text	If "Yes", supplier indicated that reported residential and/or non-residential recycled water values are preliminary values
POTABLE_SUPPLY_MINUS_SOLD_MINUS_AG_GAL	Potable Supply Minus Sold Minus Ag Gallons	Numeric	Calculated field. Potable supply minus sold water, less water used to irrigate commercially grown crops.
POTABLE_SUPPLY_MINUS_SOLD_MINUS_AG_GAL_FLAG	Potable Supply Minus Sold Minus Ag Flag	Text	If "Flagged", indicates that the corresponding Production Minus Sold Minus Ag Gallons number is an outlier in the seasonal subset of values, corresponding to a modified Z-score of ±7 or more (see Production Minus Sold Minus Ag Gallons Modified Z-Score)
POTABLE_SUPPLY_MINUS_SOLD_ZSCORE	Potable Supply Minus Sold Gallons Modified Z-Score	Numeric	Calculated modified z-score for the Potable Supply Minus Sold Gallons variable. See documentation for an explanation of the calculation procedure.
POTABLE_DEMAND_RES_ZSCORE	Potable Demand Residential Gallons Modified Z-Score	Numeric	Calculated modified z-score for the Potable Demand Residential Gallons variable. See documentation for an explanation of the calculation procedure.
R-GPCD_ZSCORE	R-GPCD Modified Z-Score	Numeric	Calculated modified z-score for the R-GPCD variable. See documentation for an explanation of the calculation procedure.
POTABLE_SUPPLY_MINUS_SOLD_MINUS_AG_ZSCORE	Potable Supply Minus Sold Minus Ag Gallons Modified Z-Score	Numeric	Calculated modified z-score for the Potable Supply Minus Sold Minus Ag Gallons variable. See documentation for an explanation of the calculation procedure.

Data dictionary for SAFER Clearinghouse per-water-system dataset

SAFER Clearinghouse Dataset Field Name	Label	Type	Description
ORG_ID	Organization ID	Text	DWR-assigned identification number associated with the urban retail water supplier
SUPPLIER_NAME	Supplier Name	Text	Name of the urban retail water supplier
WATER_SYSTEM_ID	Water System ID	Text	Identification number associated with the water system
REPORT_PERIOD_ START_DATE	Reporting Period Start Date	Timestamp	Calendar date corresponding to the start of the reporting period
REPORT_PERIOD_ END_DATE	Reporting Period End Date	Timestamp	Calendar date corresponding to the end of the reporting period
POP_REPORT_ PERIOD	Population During Reporting Period	Numeric	Estimate of the number of permanent residents served potable water during the reporting month
DWR_STANDARD_ LEVEL	DWR State Standard Level corresponding to Stage	Text	A standardized numeric level (0-6) corresponding to a supplier's local Water Shortage Contingency Plan stage
POTABLE_SUPPLY_ UOM	Potable Supply Unit of Measure	Text	Original units of reported potable supply
POTABLE_SUPPLY_OU	Potable Supply	Numeric	Total volume, in original units, of potable water produced from groundwater, GWUDI, surface water, springs, or external sources.
POTABLE_SUPPLY_ GAL	Potable Supply Gallons	Numeric	Total volume, in gallons, of potable water produced from groundwater, GWUDI, surface water, springs, or external sources.
POTABLE_SUPPLY_ PRELIM_EST	Potable Supply Preliminary Estimate	Text	If "Yes", supplier indicated that the reported potable supply number is a preliminary value
POTABLE_DEMAND_ UOM	Potable Demand Unit of Measure	Text	Original units of reported potable demand
POTABLE_VOL_SOLD_ OU	Potable Volume Sold Delivered to Other Water Systems	Numeric	Total potable volume, in original units, delivered from one supplier to other entities
POTABLE_VOL_SOLD_ GAL	Potable Volume Sold or Delivered Gallons	Numeric	Total potable volume, in gallons, delivered from one supplier to other entities
POTABLE_SUPPLY_ MINUS_SOLD_GAL	Potable Supply Minus Sold Gallons	Numeric	Total potable supply, excluding potable water delivered to other entities, in gallons

SAFER Clearinghouse Dataset Field Name	Label	Type	Description
POTABLE_DEMAND_RES_SF_OU	Potable Demand Residential Single-family	Numeric	Total potable volume, in original units, of residential single-family water use. Single-family is defined as a detached dwelling with no shared property, built on its own parcel of land.
POTABLE_DEMAND_RES_MF_OU	Potable Demand Residential Multi-family	Numeric	Total potable volume, in original units, of residential multi-family water use. A multi-family unit is a single service connection that accommodates more than one family living separately such as through a duplex, apartment, condominium, or townhouse. This includes mobile home/trailer parks that are served collectively through a master service meter.
POTABLE_DEMAND_RES_OU	Potable Demand Residential	Numeric	Sum, in original units, of single and multi-family residential water use
POTABLE_DEMAND_RES_GAL	Potable Demand Residential Gallons	Numeric	Sum, in gallons, of single and multi-family residential water use
R-GPCD	R-GPCD	Numeric	Calculated field. Residential daily per capita water use, calculated as (residential gallons)/((number of days in reporting period)×(population during reporting period))
POTABLE_DEMAND_RES_PRELIM_EST	Potable Demand Residential Preliminary Estimate	Text	If "Yes", supplier indicated that the reported residential volumes are preliminary values
POTABLE_DEMAND_AG_OU	Potable Demand Agriculture	Numeric	Total potable volume, in original units, of water used to irrigate commercially grown crops, nurseries, etc.
POTABLE_DEMAND_AG_GAL	Potable Demand Agriculture Gallons	Numeric	Total potable volume, in gallons, of water used to irrigate commercially grown crops, nurseries, etc.
POTABLE_DEMAND_CI_OU	Potable Demand Commercial Institutional	Numeric	Total potable volume, in original units, of commercial (retail, office buildings, laundries, campgrounds, gas stations, etc) and institutional (schools, prisons, hospitals, dormitories, nursing homes, hotels, etc) water use
POTABLE_DEMAND_I_OU	Potable Demand Industrial	Numeric	Total potable volume, in original units, of industrial (factories, assembly plants, and other manufacturing industries) water use

SAFER Clearinghouse Dataset Field Name	Label	Type	Description
POTABLE_DEMAND_CII_OU	Potable Demand Commercial, Industrial, and Institutional	Numeric	Potable volume of commercial, industrial, and institutional (CII) water use, in original units. Does NOT include irrigation of parks, medians, etc if said irrigation is associated with dedicated irrigation meters (DIMs).
POTABLE_DEMAND_CII_GAL	Potable Demand Commercial, Industrial, and Institutional Gallons	Numeric	Potable volume of commercial, industrial, and institutional (CII) water use, in gallons. Does NOT include irrigation of parks, medians, etc if said irrigation is associated with dedicated irrigation meters (DIMs).
POTABLE_DEMAND_IRR_OU	Potable Demand Non-Residential Metered Irrigation	Numeric	Potable volume, in original units, of CII irrigation (parks, medians, etc) associated with DIMs.
POTABLE_DEMAND_IRR_GAL	Potable Demand Non-Residential Metered Irrigation Gallons	Numeric	Potable volume, in gallons, of CII irrigation (parks, medians, etc) associated with DIMs.
POTABLE_DEMAND_O_OU	Potable Demand Other	Numeric	Potable volume of water used for purposes other than the previously reported columns, in original units. May include fire suppression, street cleaning, line flushing, construction/temporary meters, etc.
POTABLE_DEMAND_O_GAL	Potable Demand Other Gallons	Numeric	Potable volume of water used for purposes other than the previously reported columns, in gallons. May include fire suppression, street cleaning, line flushing, construction/temporary meters, etc.
POTABLE_DEMAND_PRELIM_EST	Potable Demand Preliminary Estimate	Text	If "Yes", supplier indicated that some or all of the reported potable demand fields are preliminary values
NON_POTABLE_UOM	Non-Potable Demand Unit of Measure	Text	Original units of reported non-potable demand
RES_RECYCLED_DEMAND_OU	Residential Recycled Water Demand	Numeric	Total volume, in original units, of recycled water used by both single and multi-family homes.
NON_RES_RECYCLED_DEMAND_OU	Non-Residential Recycled Water Demand	Numeric	Total volume, in original units, of non-residential recycled water use.

SAFER Clearinghouse Dataset Field Name	Label	Type	Description
RECYCLED_DEMAND_OU	Recycled Water Demand	Numeric	Calculated sum of reported residential and non-residential recycled water use, in original units. Total volume of recycled water used by the supplier for both residential and non-residential purposes during the reporting period
RECYCLED_DEMAND_GAL	Recycled Water Demand Gallons	Numeric	Calculated sum of reported residential and non-residential recycled water use, in gallons. Total volume of recycled water used by the supplier for both residential and non-residential purposes during the reporting period
NON_POTABLE_DEMAND_PRELIM_EST	Non-Potable Demand Preliminary Estimate	Text	If "Yes", supplier indicated that reported residential and/or non-residential recycled water values are preliminary values
POTABLE_SUPPLY_COMMENTS	Potable Supply Comments	Text	Caveats or additional information about the reported potable supply
POTABLE_DEMAND_COMMENTS	Potable Demand Comments	Text	Caveats or additional information about the reported potable demand
NON_POTABLE_DEMAND_COMMENTS	Non-Potable Demand Comments	Text	Caveats or additional information about the reported non-potable demand

Changes to Documentation/Dataset Methodology

- July 8, 2024: modified z-score procedure. Changed from fixed seasons to a moving 3-month window.